



DTE Platform — Product & Domain Blueprint (DTE-Type-First)



What this document is. A *product and domain* blueprint — not a technical architecture. It is detailed enough to start **database design, API design, and UI design**, but it deliberately says nothing about stacks, frameworks, deployment, or hosting. Companion to [DTE Platform — Architecture Blueprint \(El Salvador\)](#), and [DTE Platform — Domain Decomposition \(DDD\)](#).

The correction this document encodes. The prior model treated **Document** as one aggregate with a **type** attribute. Here, **DTE Type is a first-class domain concept**: a versioned specification (schema, catalogs, validations, allowed references, lifecycle peculiarities) that *governs* every document. Documents are instances projected through a type specification — not a Factura with a DTE bolted on.

Operating assumptions (given, not re-litigated): the licensee owns the infrastructure, the certificate, and the MH credentials; the licensee is the legal issuer; we ship licensed software. Legality, accreditation, cert custody, and deployment are out of scope here.

0. Framing, assumptions challenged, and ubiquitous language

0.1 Assumptions I am explicitly challenging

Stated assumption	Challenge / nuance the product must absorb
"Customer is the issuer" — one issuer per install	Accounting firms and holding groups will want one install issuing for many NITs (multi-emisor). Even single-company installs have multiple establecimientos and puntos de venta , each with independent control numbering. Model the <i>Issuer</i> (Emisor) as a first-class aggregate with multiple establishments — do not hard-code "one company."
"Platform is licensed software" → no multi-tenancy needed	Licensing does not remove tenancy: it relocates it. You still need an isolation boundary per Emisor for data, numbering, certs, permissions, and audit. Keep the tenancy seam even on a single-issuer box, or multi-emisor later becomes a rewrite.
"Customer management" is one concept	"Customer" is dangerously overloaded. Disambiguate ruthlessly (see glossary): the <i>Licensee</i> (who runs the software) vs. the <i>Receptor</i> (the buyer on a DTE) vs. the <i>Comercial Counterparty</i> (an AR/AP relationship). Conflating them corrupts the whole model.
The platform only <i>issues</i> documents	A real fiscal product must also receive DTEs (supplier documents), because IVA crédito fiscal, retención, and settlement workflows depend on inbound documents. "DTE generation" is only half the domain. (See §12, Missing Modules.)
"Substitution" is a small feature	Substitution is a two-document saga (invalidate original referencing the replacement + issue replacement) with strict ordering and its own failure modes. It deserves its own process aggregate, not a checkbox.
"Payment tracking" = marking invoices paid	Payments touch the fiscal layer (condición de operación, forma de pago on the DTE) <i>and</i> a commercial layer (AR ledger, partial payments, settlement). These are different bounded contexts with a one-way dependency. Modeling them as one creates the classic invoicing-monolith mess.

0.2 Ubiquitous language (the words we will not blur)

Term	Means in this platform	Never confused with
Licensee	The organization that runs the software and owns the infra/cert/credentials.	The Receptor.
Emisor (Issuer)	The legal taxpayer (NIT/NRC) on whose behalf DTEs are issued. One Licensee may operate several.	Establecimiento / POS.
Establecimiento	A registered place of business of an Emisor (has its own MH code).	Punto de Venta.

Term	Means in this platform	Never confused with
Punto de Venta (POS)	A logical issuance point inside an establecimiento; drives control-number sequencing.	A physical device.
Receptor	The counterparty named <i>on a specific DTE</i> (buyer). A snapshot, not a live link.	Comercial Counterparty.
Comercial Counterparty	A persistent business relationship in the AR/AP ledger (a "client" or "supplier").	Receptor snapshot.
DTE Type	A first-class, versioned <i>specification</i> : schema + catalogs + validations + allowed references + lifecycle rules.	A document instance.
Fiscal Document	An instance governed by a DTE Type specification.	The PDF/visual representation.
Sello de Recepción	MH's acceptance stamp; the legal proof of reception.	The JWS signature.
Anulación	Invalidation of an already-accepted DTE (MH event with its own types).	Discarding a draft.
Substitution	Invalidate-and-replace: an anulación (tipo 1) bound to a newly issued replacement.	Nota de Crédito/Débito.

1. The central modeling decision — DTE Type as a first-class concept

1.1 Why the type must be first-class

In the Salvadoran regime, structure is dictated *by Hacienda*: each tipo de DTE has its own JSON schema, its own required catalogs, its own validation rules, its own reference obligations, and its own lifecycle quirks. Modeling a single **Factura** with variations is backwards. Instead:

- A **DocumentTypeSpecification** is a domain object the platform *knows about and reasons over*.
- A **FiscalDocument** is always created **through** a specification — the spec decides what fields exist, what is required, what references are mandatory, who can be a valid receptor, and which lifecycle transitions are legal.
- Adding a future DTE type = registering a new specification (schema + rules + lifecycle), **not** rewriting the document model.



Design rule: **No code path may hard-code a tipo de DTE**. Type-specific behavior lives in the specification (data + rules), and the engine is generic. The only place that "knows" FE differs from CCF is the registry of specifications.

1.2 The DTE Type catalog (first-class registry)

Each row is a **DocumentTypeSpecification** (versioned). Codes follow MH's tipo de documento.

Code	DTE Type	Primary purpose	Receptor profile	Mandatory references	Notable lifecycle quirk
01	Factura (FE)	Sale to final consumer	Consumer (DUI/optional)	None	Correctable via NC/ND or substitution
03	Comprobante de Crédito Fiscal (CCF)	Sale to IVA contribuyente	Contribuyente (NIT+NRC required)	None	Basis for CR, NC, ND
04	Nota de Remisión (NR)	Goods movement / delivery	Either	Often references sale	Not a value-settling doc
05	Nota de Crédito (NC)	Reduce value of a prior CCF	Same as referenced	Must reference an accepted CCF	Cannot exceed referenced amounts
06	Nota de Débito (ND)	Increase value of a prior CCF	Same as referenced	Must reference an accepted CCF	Mirror of NC
07	Comprobante de Retención (CR)	Document IVA withholding	Contribuyente	References withheld CCF(s)	Issued by withholding agent
08	Comprobante de Liquidación (CL)	Settle operations on behalf of a third party	Principal	References settled documents	Drives settlement workflow

Code	DTE Type	Primary purpose	Receptor profile	Mandatory references	Notable lifecycle quirk
09	Documento Contable de Liquidación (CD)	Accounting liquidation record	Principal	References period/docs	Periodic, aggregate
11	Factura de Exportación (FEX)	Export sale	Foreign receptor	None	Incoterms, foreign-ID rules
14	Factura de Sujeto Excluido (FSE)	Purchase from an excluded subject	Excluded subject (as "seller")	None	Issuer documents an inbound purchase — inverted role
15	Comprobante de Donación (CD-Don)	Donation receipt	Donor	None	Non-commercial totals
...	Future types	Whatever MH adds	—	—	Register a new spec; no rewrite



FSE is the assumption-breaker. In a Factura-de-Sujeto-Excluido the *issuer* documents money it is *paying out* to an excluded subject. If your model assumes "document = something I sold," FSE breaks it. Treat "flow direction" (outbound sale vs. inbound-on-behalf) as a property of the **type specification**, not an afterthought.

1.3 Anatomy of a **DocumentTypeSpecification**

- **Identity:** `typeCode` (MH), `schemaVersion`, `effectiveFrom`, `effectiveTo`, `status` (`draft` / `active` / `deprecated`).
- **Structure contract:** field set, required/optional, repeatable sections (lines, taxes, references, payments), allowed catalogs (units, tax codes, payment forms, operation conditions).
- **Validation rules:** declarative invariants the engine evaluates (e.g., "CCF requires receptor NRC", "NC total ≤ referenced document remaining balance", "FEX requires Incoterm").
- **Reference policy:** which target types are allowed/required, and constraints on their state (e.g., "reference must be `ACCEPTED`").
- **Receptor policy:** who is a legal receptor (consumer, contribuyente, foreign, excluded subject).
- **Lifecycle overlay:** type-specific allowed transitions and windows (e.g., anulación window, backdating window for contingencia).
- **Numbering policy:** how the control number series is keyed for this type.
- **Flow direction:** `OUTBOUND_SALE` / `INBOUND_ON_BEHALF` / `MOVEMENT` / `ADJUSTMENT` / `WITHHOLDING` / `SETTLEMENT`.

This object is what UI form-builders, API validators, and DB schema generators all read from. It is the single source of truth for "what a document of this type *is*."

2. Bounded contexts (product view)

The DDD page lists technical contexts; here is the **product-capability map**, with two contexts the prior model lacked (Type Registry, Receivables/Settlement) and one renamed for clarity (Reception).

```

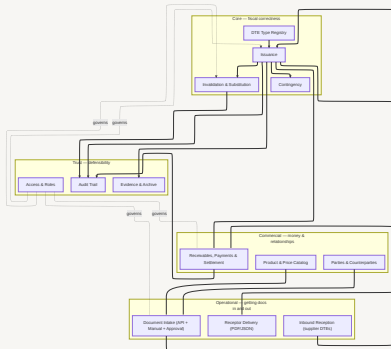
flowchart TB
  subgraph CORE["Core – fiscal correctness"]
    TYPES["DTE Type Registry"]
    ISS["Issuance"]
    INV["Invalidation & Substitution"]
    CONT["Contingency"]
  end
  subgraph COMMERCE["Commercial – money & relationships"]
    PARTY["Parties & Counterparties"]
    CAT["Product & Price Catalog"]
    AR["Receivables, Payments & Settlement"]
  end
  subgraph FLOW["Operational – getting docs in and out"]
    INTAKE["Document Intake (API + Manual + Approval)"]
  end

```

```

RECEP["Inbound Reception (supplier DTEs)"]
DELIV["Receptor Delivery (PDF/JSON)"]
end
subgraph TRUST["Trust – defensibility"]
  EVID["Evidence & Archive"]
  AUD["Audit Trail"]
  ACCESS["Access & Roles"]
end
end
TYPES --> ISS
INTAKE --> ISS
CAT --> INTAKE
PARTY --> INTAKE
ISS --> INV
ISS --> CONT
ISS --> AR
ISS --> DELIV
ISS --> EVID
RECEP --> AR
ISS --> AUD
INV --> AUD
AR --> AUD
ACCESS -.governs.-> ISS & INTAKE & INV & AR

```



Context	Product responsibility	Why it is its own context
DTE Type Registry	Owns type specifications, schema versions, catalogs, validation rule sets.	Types change on MH's schedule, not yours; isolating them lets you onboard a new type as data.
Issuance	Builds, validates, numbers, signs, transmits, and tracks acceptance of fiscal documents of any type.	The fiscally binding core. Bugs here are legal liability.
Invalidation & Substitution	Anulación (with MH event types) and the invalidate-and-replace saga.	It is a multi-document process with its own states and failure modes.
Contingency	Episodes, enrollment, batch resubmission, contingency-type classification.	A first-class legal regime, not error handling.
Parties & Counterparties	Persistent records of clients/suppliers, tax identities, contacts, default fiscal profiles.	Live, mutable data — must be <i>snapshotted</i> into documents, never linked live.
Product & Price Catalog	Goods/services, units, tax treatment, price lists.	Churns daily; documents must freeze its values.
Receivables, Payments & Settlement	AR ledger, payment application, partial payments, statements, liquidation/settlement runs.	Commercial money truth — distinct from fiscal truth; depends on Issuance one-way.
Document Intake	API issuance, manual issuance, and the approval workflow for manual/exception documents.	The human + machine front door with its own governance and draft lifecycle.
Inbound Reception	Capture and validate supplier DTEs received by the issuer.	Needed for IVA crédito, retención, and settlement; a different flow direction.
Receptor Delivery	Generate and deliver the human representation (PDF/JSON) to receptors.	Aesthetic + delivery concern; must never sit in the issuance hot path.
Evidence & Archive	Immutable preservation of signed payloads, MH responses, and reconstructable representations.	Regulatory deliverable; must outlive every rewrite.
Audit Trail	Append-only, tamper-evident record of every meaningful action and transition.	Defensibility; cannot be backfilled.

Context	Product responsibility	Why it is its own context
Access & Roles	Users, roles, permissions, service accounts, approval authority.	Cross-cutting governance; drives every workflow guard.

3. Aggregate roots (per context)

Context	Aggregate root	Guards (consistency boundary)
Type Registry	DocumentTypeSpecification	One coherent, versioned spec; activation/deprecation transitions.
Type Registry	Catalog (units, tax codes, payment forms, operation conditions, departments/municipalities, countries, Incoterms)	Effective-dated reference values; non-overlapping windows.
Issuance	FiscalDocument	Immutability after signing; totals consistency; reference correctness; receptor compatibility.
Issuance	ControlNumberSeries	Monotonic, gapless allocation per (Emisor, type, establecimiento, POS, ambiente).
Issuance	TransmissionRecord	Append-only attempt history; acceptance requires a sello.
Invalidation	InvalidationEvent	Only over an accepted doc; anulaci3n type rules; window enforcement.
Invalidation	SubstitutionProcess (saga)	Ordering: replacement issued + original invalidated referencing it; compensations.
Contingency	ContingencyEpisode	One open episode per cause; batch closure rules; legal window flagging.
Parties	Counterparty	Tax-identity validity; profile completeness per intended document types.
Catalog	Item (Product/Service)	Price ≥ 0; valid unit/tax code; active state.
Catalog	PriceList	Currency consistency; effective windows.
Receivables	ReceivableAccount (per counterparty)	Balance = Σ posted invoices – Σ applied payments/credits.
Receivables	PaymentReceipt	Cannot over-apply beyond document balances; currency match.
Receivables	SettlementRun	Closes a period/relationship; references settled documents exactly once.
Intake	IssuanceRequest (draft + approval envelope)	Source (API/manual); approval state; converts to exactly one FiscalDocument.
Reception	ReceivedDocument	Authenticity (validate MH sello/signature); dedupe by c3digo de generaci3n.
Delivery	DeliveryJob	Channel + status; retries; idempotent per (document, channel, recipient).
Evidence	EvidenceBundle	Write-once; integrity hash; completeness (payload + sello + representation).
Audit	AuditEntry	Append-only hash chain per Emisor.
Access	User, ServiceAccount, Role	At least one owner; revocation is synchronous; scoped credentials.

4. Domain entities (inside aggregates)

- **FiscalDocument** → **DocumentLine**, **TaxAllocation**, **DocumentReference**, **PaymentTerm**, **Observation**, **RelatedDocsBlock** (type-specific repeatable sections defined by the spec).

- **DocumentTypeSpecification** → `FieldDefinition`, `ValidationRule`, `ReferenceRule`, `ReceptorRule`, `LifecycleRule`, `SchemaVersionBinding`.
- **Counterparty** → `TaxIdentity`, `ContactPerson`, `FiscalProfile` (per role: as-client / as-supplier), `Address`.
- **Item** → `UnitBinding`, `TaxTreatment`, `PriceListEntry`.
- **ReceivableAccount** → `LedgerPosting` (invoice posted, payment applied, credit applied, adjustment).
- **PaymentReceipt** → `PaymentAllocation` (links a payment to one or more documents), `PaymentInstrument`.
- **SettlementRun** → `SettlementLine` (document settled), `SettlementOutputRef` (e.g., CL/CD issued).
- **IssuanceRequest** → `DraftPayload`, `ApprovalStep`, `ApprovalDecision`, `ValidationFinding`.
- **ContingencyEpisode** → `EnrolledDocument`.
- **TransmissionRecord** → `TransmissionAttempt`.
- **ReceivedDocument** → `ReceivedLine`, `WithholdingDetail`.
- **EvidenceBundle** → `EvidenceArtifact` (signed payload / MH response / rendered representation / hash manifest).

5. Value objects (immutable)

Group	Value objects
Fiscal identity	<code>CodigoGeneracion</code> (UUID), <code>NumeroControl</code> (<code>DTE-XX-XXXXXXXX-NNNNNNNNNNNN</code>), <code>SelloRecepcion</code> , <code>TipoDte</code> , <code>SchemaVersion</code> , <code>Ambiente</code> (test/prod)
Tax identity	<code>Nit</code> , <code>Nrc</code> , <code>Dui</code> , <code>ForeignTaxId</code> , <code>TaxpayerKind</code> (consumer / contribuyente / excluded / foreign), <code>EconomicActivityCode</code>
Money & tax	<code>Money</code> (decimal + currency), <code>TaxRate</code> , <code>TaxCode</code> , <code>Totals</code> (gravado, exento, no sujeto, IVA, IVA retenido/percibido, total), <code>RetentionAmount</code>
Commercial terms	<code>OperationCondition</code> (contado / crédito / otro), <code>PaymentForm</code> (efectivo, transferencia, tarjeta, ...), <code>DueDate</code> , <code>Incoterm</code>
Snapshots	<code>ReceptorSnapshot</code> , <code>EmisorSnapshot</code> , <code>ItemSnapshot</code> — frozen at validation/signing time
References	<code>DocumentReferenceV0</code> (target código de generación + type + reason), <code>SubstitutionLink</code> , <code>WithheldDocumentRef</code>
Lifecycle	<code>DocumentStatus</code> , <code>InvalidationType</code> (1 corrección, 2 rescindir, 3 otro), <code>ContingencyType</code> (MH-defined causes), <code>TransmissionOutcome</code>
Time & place	<code>IssuedAt</code> , <code>FiscalPeriod</code> , <code>Establecimiento</code> , <code>PuntoVenta</code> , <code>GeoLocation</code>



Snapshot law. Counterparty, Emisor, and Item data are *live* in their contexts but become **immutable value objects inside a document** at validation time. A document is a photograph of the facts at issuance, never a live join. This is the single most important rule for fiscal correctness.

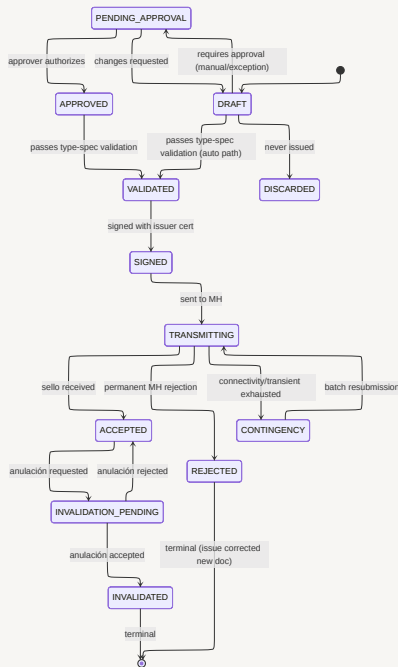
6. Lifecycle states

6.1 Generic issuance lifecycle (governed by the type spec)

Every FiscalDocument runs this machine; the **type specification supplies the guards** (which transitions are legal, what windows apply).

```
stateDiagram-v2
    [*] --> DRAFT
    DRAFT --> PENDING_APPROVAL: requires approval (manual/exception)
    PENDING_APPROVAL --> DRAFT: changes requested
    PENDING_APPROVAL --> APPROVED: approver authorizes
    DRAFT --> VALIDATED: passes type-spec validation (auto path)
    APPROVED --> VALIDATED: passes type-spec validation
    VALIDATED --> SIGNED: signed with issuer cert
```

SIGNED --> TRANSMITTING: sent to MH
TRANSMITTING --> ACCEPTED: sello received
TRANSMITTING --> REJECTED: permanent MH rejection
TRANSMITTING --> CONTINGENCY: connectivity/transient exhausted
CONTINGENCY --> TRANSMITTING: batch resubmission
ACCEPTED --> INVALIDATION_PENDING: anulaci3n requested
INVALIDATION_PENDING --> INVALIDATED: anulaci3n accepted
INVALIDATION_PENDING --> ACCEPTED: anulaci3n rejected
REJECTED --> [*]: terminal (issue corrected new doc)
INVALIDATED --> [*]: terminal
DRAFT --> DISCARDED: never issued



6.2 Type-specific overlays (examples)

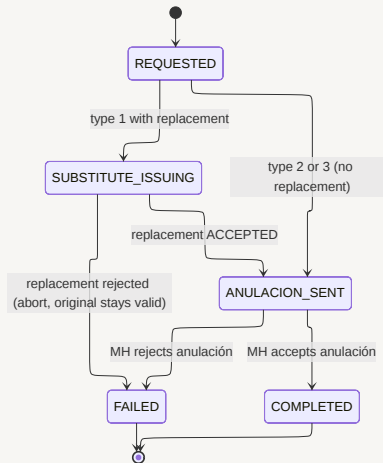
- **NC/ND (05 / 06)**: cannot enter **VALIDATED** unless a referenced **accepted CCF** exists and the adjustment respects remaining balance.
- **CR (07)**: **VALIDATED** requires at least one withheld document reference and a computed retention amount.
- **FEX (11)**: requires Incoterm + foreign receptor identity; consumer-receptor rules disabled.
- **FSE (14)**: flow is **INBOUND_ON_BEHALF**; the "receptor" is the excluded subject being paid; feeds AP, not AR.
- **CL/CD (08 / 09)**: created by a **SettlementRun**, not by ad-hoc manual entry.

6.3 Invalidation & substitution lifecycle

```

stateDiagram-v2
    [*] --> REQUESTED
    REQUESTED --> SUBSTITUTE_ISSUING: type 1 with replacement
    SUBSTITUTE_ISSUING --> ANULACION_SENT: replacement ACCEPTED
    REQUESTED --> ANULACION_SENT: type 2 or 3 (no replacement)
    ANULACION_SENT --> COMPLETED: MH accepts anulaci3n
    ANULACION_SENT --> FAILED: MH rejects anulaci3n
    SUBSTITUTE_ISSUING --> FAILED: replacement rejected (abort, original stays valid)
    COMPLETED --> [*]
    FAILED --> [*]

```

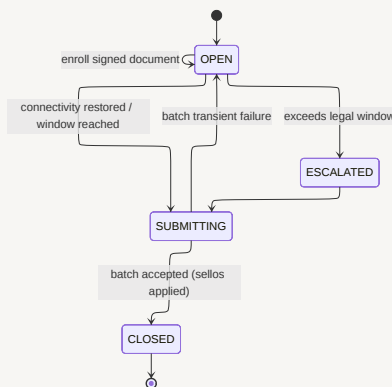


Substitution ordering is non-negotiable. The replacement must reach **ACCEPTED** **before** the anulaci3n tipo 1 that references it is sent. If the replacement fails, you must *not* invalidate the original. Model this as a saga with explicit compensation, never as two independent button presses.

6.4 Contingency episode lifecycle

```

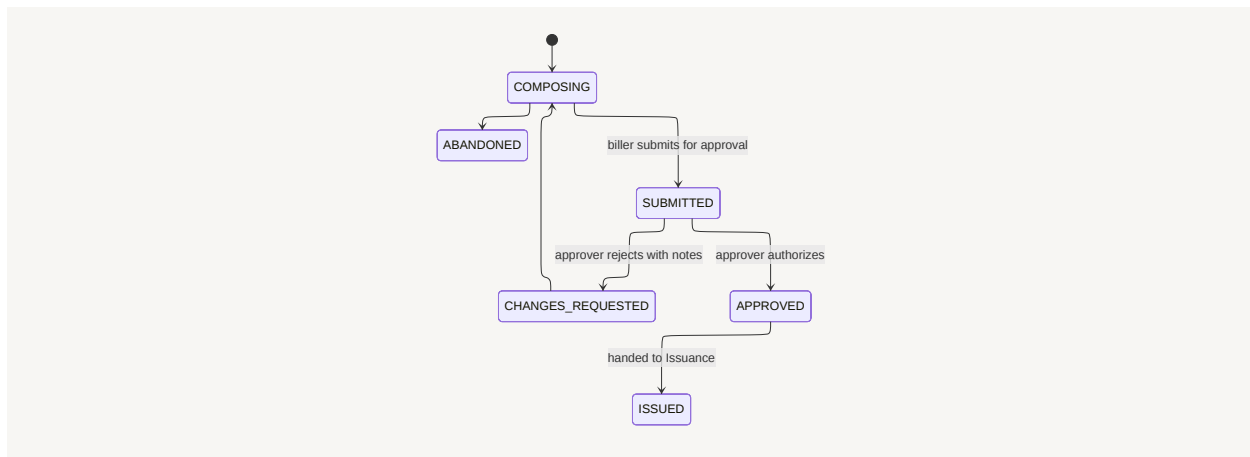
stateDiagram-v2
    [*] --> OPEN
    OPEN --> OPEN: enroll signed document
    OPEN --> SUBMITTING: connectivity restored / window reached
    SUBMITTING --> OPEN: batch transient failure
    SUBMITTING --> CLOSED: batch accepted (sellos applied)
    OPEN --> ESCALATED: exceeds legal window
    ESCALATED --> SUBMITTING
    CLOSED --> [*]
  
```



6.5 Manual approval lifecycle (Intake)

```

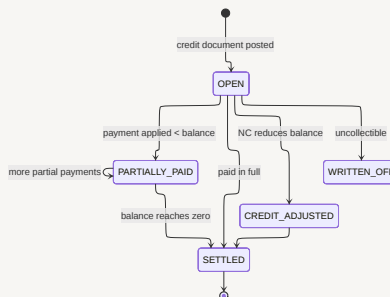
stateDiagram-v2
    [*] --> COMPOSING
    COMPOSING --> SUBMITTED: biller submits for approval
    SUBMITTED --> CHANGES_REQUESTED: approver rejects with notes
    CHANGES_REQUESTED --> COMPOSING
    SUBMITTED --> APPROVED: approver authorizes
    APPROVED --> ISSUED: handed to Issuance
    COMPOSING --> ABANDONED
  
```

6.6 Receivable / payment lifecycle (commercial)

```

stateDiagram-v2
    [*] --> OPEN: credit document posted
    OPEN --> PARTIALLY_PAID: payment applied < balance
    PARTIALLY_PAID --> PARTIALLY_PAID: more partial payments
    PARTIALLY_PAID --> SETTLED: balance reaches zero
    OPEN --> SETTLED: paid in full
    OPEN --> CREDIT_ADJUSTED: NC reduces balance
    CREDIT_ADJUSTED --> SETTLED
    SETTLED --> [*]
    OPEN --> WRITTEN_OFF: uncollectible
  
```



7. Business workflows

7.1 API-based issuance (machine path)

1. Integrator calls *issue* with an idempotency key and a type code.
2. **Type Registry** resolves the active specification + schema version.
3. **Intake** builds an `IssuanceRequest`; the spec drives validation. Approval is skipped if the integration is trusted for that type (policy).
4. **Issuance** allocates the control number, snapshots receptor/items, signs, transmits.
5. Acceptance returns a sello; events fire to Receivables (if credit), Delivery, Evidence, Audit.
6. Response returns document identity + status URL; the caller polls or receives a webhook.

7.2 Manual issuance with approval (human path)

1. Biller composes a document in a type-aware form (fields rendered from the spec).
2. For exception cases (manual override, high value, backdated, unusual receptor), policy routes it to `PENDING_APPROVAL`.

3. An **Approver** reviews findings, then approves or requests changes.
4. On approval, the same Issuance engine takes over — *identical* downstream path to API issuance.



Approval is an Intake concern, not an Issuance concern. Issuance must not know whether a document came from API or human, approved or auto. It receives a *validated, authorized request*. This keeps the fiscal core uniform and testable.

7.3 Invalidation (anulación)

- Triggered only over an **ACCEPTED** document, within the MH-allowed window.
- Capture **InvalidationType** (1 corrección con reemplazo, 2 rescindir operación, 3 otros) and a reason.
- Type 1 demands a replacement (→ substitution saga). Types 2/3 stand alone.

7.4 Substitution (invalidate-and-replace)

- A saga: issue replacement → on acceptance, send anulación tipo 1 referencing the replacement's código de generación → on acceptance, mark original **INVALIDATED** and link **SubstitutionLink** both ways.
- Receivables must move balances from the original to the replacement atomically at the commercial layer.

7.5 Contingency

- On transient transmission failure exhaustion, enroll signed docs into an open **ContingencyEpisode** with a **ContingencyType**.
- When connectivity returns or the window forces it, submit the batch; apply sellos per document; close the episode. Episodes beyond the legal window are escalated for compliance review.

7.6 Payment tracking

- When a credit (**crédito**) document is accepted, post a **ReceivableAccount** entry.
- Record **PaymentReceipt**s and apply them (full or partial) to one or more documents; the account balance and document status update.
- NC/ND adjust balances via references, not by editing documents.

7.7 Invoice settlement / liquidation

- A **SettlementRun** closes a relationship or period: it gathers settled documents, reconciles applied payments and adjustments, and may produce a **Comprobante de Liquidación (CL)** or **Documento Contable de Liquidación (CD)** through Issuance.
- Output: a statement plus, where applicable, a fiscal liquidation document — each settled document referenced exactly once.

7.8 Inbound reception (supplier DTEs)

- Capture supplier DTEs (upload, email intake, or API), validate their MH sello/signature, dedupe by código de generación.
- Feed IVA crédito fiscal records, support retención (CR issuance references them), and provide AP/settlement inputs.

7.9 Receptor delivery

- After acceptance, generate the human representation and deliver via the configured channel(s); retries are idempotent and never block issuance.

8. Roles & permissions (product matrix)

Roles are coarse on purpose; refine only when real need appears. Permissions are checked at **workflow guards**, expressed in product terms.

Capability	Owner	Fiscal Mgr	Billor	Approver	Accountant	Auditor (RO)	Service Ac (API)
Manage Emisor / establishments / POS	✓	✓	—	—	—	—	—
Manage users & roles	✓	—	—	—	—	—	—
Manage catalog & counterparties	✓	✓	✓	—	✓	—	(scoped)
Compose / submit documents	✓	✓	✓	—	—	—	✓
Approve manual/exception docs	✓	✓	—	✓	—	—	—
Issue / transmit	✓	✓	✓ (within limits)	—	—	—	✓
Invalidate / substitute	✓	✓	—	(approve)	—	—	(scoped)
Record payments / run settlement	✓	✓	—	—	✓	—	(scoped)
Manage certificates / MH credentials	✓	✓	—	—	—	—	—
View audit & export evidence	✓	✓	—	—	✓	✓	—

Authority dimensions beyond role: value thresholds (e.g., approval required above an amount), document-type scoping (a service account limited to FE only), establishment scoping, and ambiente (test vs prod). These are *policies*, layered on top of roles.

9. Audit & evidence preservation requirements

Requirement	Product behavior
Every state transition is recorded	Actor (user/service/system), timestamp, before/after status, reason — append-only.
Tamper-evidence	Hash-chained audit entries per Emisor; a verification routine can prove the chain is intact.
Evidence completeness	Each accepted document has an EvidenceBundle : signed payload + MH response/sello + reconstructable representation + integrity manifest.
Source of legal truth	The signed payload (not the DB row) is canonical; disputes are answered from the EvidenceBundle.
Inspection export	One action produces a complete, verifiable export for a period or document set; the export itself is audited.
Approval traceability	Who composed, who approved, what findings existed, and what was overridden — preserved with the document.
Retention	Evidence is preserved for the full legal retention period and is write-once.
Sensitive-data discipline	Credentials, cert material, and secrets never appear in audit payloads or representations.

10. Event model (domain events)

Naming: `<context>.<aggregate>.<pastTenseFact>`. Every event carries emisor identity, occurredAt, actor, and a version.

Context	Key events
Type Registry	<code>type.spec.registered</code> , <code>type.spec.activated</code> , <code>type.spec.deprecated</code> , <code>catalog.updated</code>
Intake	<code>intake.request.composed</code> , <code>intake.request.submitted</code> , <code>intake.request.changes_requested</code> , <code>intake.request.approved</code> , <code>intake.request.abandoned</code>
Issuance	<code>doc.drafted</code> , <code>doc.validated</code> , <code>doc.number_allocated</code> , <code>doc.signed</code> , <code>doc.transmission_started</code> , <code>doc.accepted</code> , <code>doc.rejected</code> , <code>doc.enrolled_in_contingency</code>
Invalidation	<code>doc.invalidation_requested</code> , <code>doc.invalidated</code> , <code>doc.substituted</code> (<i>carries both códigos de generación</i>)
Contingency	<code>contingency.opened</code> , <code>contingency.document_enrolled</code> , <code>contingency.batch_submitted</code> , <code>contingency.closed</code> , <code>contingency.escalated</code>
Reception	<code>inbound.document_received</code> , <code>inbound.document_validated</code> , <code>inbound.duplicate_detected</code>
Receivables	<code>receivable.opened</code> , <code>payment.recorded</code> , <code>payment.applied</code> , <code>receivable.settled</code> , <code>receivable.written_off</code> , <code>settlement.run_completed</code>
Delivery	<code>delivery.requested</code> , <code>delivery.succeeded</code> , <code>delivery.failed</code>
Access	<code>user.invited</code> , <code>user.disabled</code> , <code>role.assigned</code> , <code>service_account.created</code> , <code>service_account.revoked</code>

Delivery semantics (product-level): Audit consumes everything synchronously (auditing a fact is part of committing it). Receivables, Delivery, and Reception consume asynchronously and must be idempotent. Webhooks to integrators are a curated subset of these events.

11. Integration boundaries

Boundary	Direction	Contract owned by	Product rule
MH (Hacienda)	Out/in	External authority	Shielded behind an anti-corruption layer; internal vocabulary never mirrors MH wire fields.
Issuer's ERP/POS	In	Our public issuance API	Stable, versioned, type-aware contract; idempotency mandatory.
Webhooks to integrators	Out	Our event contract	Signed, replay-protected, at-least-once; consumers dedupe by event id.
Receptor delivery (email/portal)	Out	Delivery context	Never in the issuance hot path; failures don't affect fiscal status.
Inbound supplier DTEs	In	Reception context	Authenticity validated before acceptance; deduped by código de generación.
Accounting/tax reporting	Out	Reporting (future)	Read-side projections (Libro de Ventas/Compras, retention summaries).
Certificate / MH credentials	—	Licensee-owned	The platform <i>uses</i> a signing capability; it never exposes raw material.

12. Missing modules identified (not in the prior model)

1. **DTE Type Registry** — the first-class type specification engine. Without it, every new MH type is a code change. (*Now core.*)
2. **Inbound Reception** — supplier DTE capture. Required for IVA crédito, retención (CR references received CCFs), and settlement. The prior model is issue-only.
3. **Receivables / Payments / Settlement** — commercial money truth, partial payments, statements, CL/CD liquidation runs.
4. **Intake & Approval** — the manual-issuance governance layer (compose → approve → issue) distinct from the fiscal core.
5. **Reporting & Fiscal Books (near-term future)** — Libro de Ventas / Compras, retention and perception summaries, period closes. Read-only projections, but a real product surface.
6. **Counterparty management as a context** — separated from the receptor *snapshot*; persistent client/supplier records with per-role fiscal profiles.

7. **Evidence & Archive as a product surface** — inspection-ready bundles and exports, not just a storage detail.
8. **Notifications to issuer staff** — cert expiry, rejected documents, stuck contingency, approval queues (operational, human-facing).

13. Future scalability concerns

Concern	Why it bites later	Design seam to protect now
High-volume POS issuance	Control-number allocation per (type, establecimiento, POS) becomes the contention point.	Keep series allocation a tiny, isolated aggregate; key it finely so contention is per-POS, not global.
MH adds/changes types & schemas	Hard-coded type logic forces redeploys and risks historical doc readability.	Type Registry + schema versioning; store the raw signed payload as the canonical artifact.
Multi-emisor (accounting firms)	Single-issuer assumptions leak into numbering, certs, permissions, audit.	Emisor as a first-class tenancy boundary from day one, even on single-issuer installs.
10-year evidence growth	Evidence volume dwarfs operational data over years.	Separate hot operational state from cold evidence; design EvidenceBundle for archival tiers.
Reporting load on the issuance store	Fiscal-book queries compete with issuance throughput.	Treat reporting as read-side projections, never as live joins over the issuance core.
Batch operations	Bulk import, bulk invalidation, mass settlement appear once volume grows.	Model batches as their own processes with per-item status, not loops over single-item APIs.
Settlement complexity	Commission/third-party liquidation (CL/CD) can become intricate.	Keep SettlementRun a dedicated aggregate that <i>consumes</i> fiscal events, never embeds fiscal logic.

14. Open decisions for the team (resolve before DB/API freeze)

1. **Multi-emisor now or seam-only?** Ship single-emisor UX but keep the Emisor boundary — confirm.
2. **Inbound reception scope for v1:** full IVA-crédito support, or capture-and-store only?
3. **Approval policy model:** fixed rules (value/type/backdate) vs. configurable rule builder.
4. **Settlement outputs:** which liquidation documents (CL 08, CD 09) are in scope for v1.
5. **Payment layer depth:** lightweight "paid/unpaid" vs. full partial-payment AR ledger.
6. **Receptor portal:** delivery only, or a receptor self-service view of their documents.
7. **Reporting in v1:** Libro de Ventas/Compras now, or post-MVP.



One-line summary: Make the **DTE Type** a first-class, versioned specification that governs a generic document engine; separate **fiscal truth** (Issuance, Invalidation, Contingency, Evidence) from **commercial truth** (Counterparties, Catalog, Receivables/Settlement) and from **operational flow** (Intake/Approval, Reception, Delivery); govern it all with role + policy guards and an append-only, tamper-evident audit — so the product can absorb new document types, multi-emisor, payments, and settlement without ever rewriting the core.